

BMI Lie Detector: Can Lifestyle Habits Predict Diabetes Risk Better than BMI Alone?

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Introduction

Body Mass Index (BMI) is a common way used in healthcare to group people into categories like underweight, normal, and overweight. It uses the formula weight (kg) divided by height squared (m^2). It is often used as a quick indicator of the health of the person. However, BMI does not take into account other important factors such as sleep habits, physical activity, stress levels, and heart health. Because of this, people with the same BMI can have a very different lifestyle and health conditions.

The motivation behind this project is to investigate whether lifestyle-related factors can predict diabetes risk better than BMI alone. The goal is to evaluate whether a variety of health and behavior features provides stronger predictive power than relying on single measurement.

Dataset And Features

We used the Health Risk Predictor Dataset from Kaggle, which contains information on 97,297 people. The dataset originally had 33 columns, but we removed pre-generated risk scores and that left us with 27 useful columns. Our target variable was whether a person's HbA1c level was 6.5% or higher which is the World Health Organization official diabetes threshold. We tried to make it a medically meaningful target. The target variables were nearly balanced and no oversampling or undersampling techniques were required.

Groups	Count	Percentage
At Risk	49,705	51.1%
Not at Risk	47,592	48.9%

Model A (control)

Model A uses a single feature: BMI

This represents the traditional approach of using BMI as a standalone health indicator.

Model B (experimental)

Model B uses 11 features that capture a broader picture of a person's lifestyle and health:

- Age
- Gender
- Physical Activity (minutes per week)
- Diet Score
- Hours of Sleep per Day
- Smoking Status
- Alcohol Consumption per Week
- Hypertension History
- Cardiovascular History
- Waist to Hip Ratio
- Screen Time Hours per Day

We removed family history, and bmi because it gave model B an unfair advantage and made the comparison less honest.

Data Preprocessing

Before training, we cleaned and prepared the data in several ways:

- **Label Encoding:** The text based columns were converted to numbers.
- **Feature Scaling:** For features on very different scales. We used separated scalers for Model A and Model B, and each scaler was fitted only on the training data to prevent data leakage.
- **Dropped Risk Scores:** The dataset included pre-generated risk scores and they were removed to keep our results honest and reliable.

Benchmark Models

We tested three classic machine learning models as benchmarks. These are commonly used in healthcare research and give a good baseline for comparison.

Logistic Regression

It's a widely used classification algorithm in medical research. We set `max_iter=1000` to make sure the model had enough time to fully converge the large dataset size.

Random Forest

This is useful because diabetes risk likely depends on combinations of habits, not just one factor at a time. We used 100 trees, a max depth of 5, and a minimum of 10 samples per leaf. We choose this setting because we want to reduce overfitting while still capturing meaningful patterns.

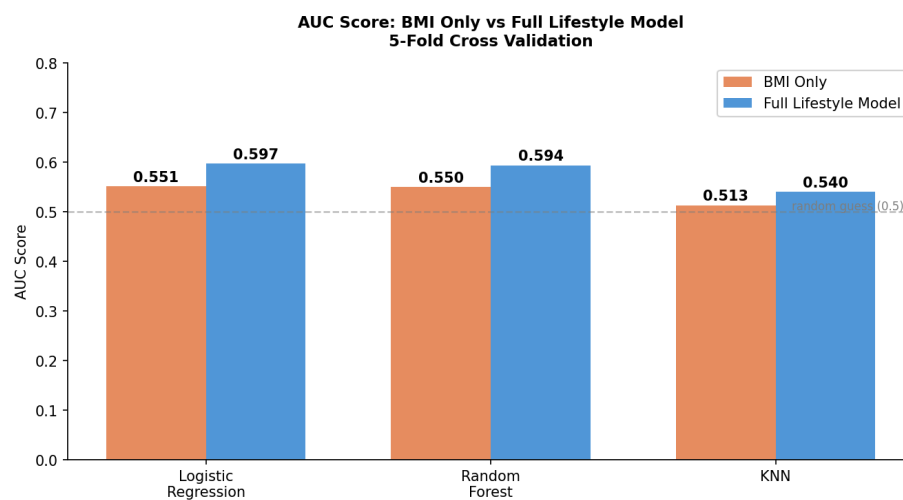
K-Nearest Neighbors (KNN)

KNN predicts a person's risk by looking at the k most similar people in the dataset and seeing what their outcomes were. We used k=9 neighbors to avoid overfitting and appropriate for a dataset of this size.

Evaluation

All models were evaluated using Stratified 5-fold cross validation. Each fold kept the same class ratio as the full dataset. Our primary metric was AUC which measures how well a model can distinguish between the two classes.

Results



Model	AUC-BMI only	AUC- Full Lifestyle	Improvement
Logistic Regression	0.551	0.597	+0.046
Random Forest	0.550	0.594	+0.044
K-Nearest Neighbors	0.513	0.540	+0.027

The average AUC improvement across all three models was +0.039. Logistic Regression was the best performing model overall, achieving an AUC of 0.0597 with the full lifestyle features.

Project Challenges

Initially, our original idea was to make a custom “health score” and compare it against the BMI. The problem was that we had to define that score ourselves because of the dataset we had. This meant we were building the answer into the data before the model was trained and didn’t have a good use of machine learning. That came more into light when we did our poster. It was a circular problem.

After that didn’t work, we wanted to find a better dataset with a similar variety of features. We looked for a target variable completely independent from our features. That’s how we landed in this diabetes risk idea. Getting to that decision took a few rounds of going back and forth but it led to a much more meaningful project that aligned more with our machine learning content.

Conclusion

The BMI-only model achieved an (AUC = 0.55) barely above random guessing (0.50). The full lifestyle model did better across all three algorithms, with the best result reaching AUC of 0.597. While the gap is small, it is consistent. The AUC values are not dramatically high, but that was expected. We choose to exclude family history of diabetes, which would have been one of the strongest predictors, to focus more on a person’s lifestyle.

What surprised us was how close BMI came to the lifestyle model despite being a single measurement. BMI is not bad too but just an incomplete one. Even a small set of lifestyle features can do better.

Contribution

Vanessa Bagolor: Logistic regression, Chart 1, Introduction report, Benchmark report, Dataset & Feature, Challenges

Shruti Dhimal: Random Forest, KNN, Chart 2, Data processing, Benchmark Report, Result, Conclusion

From the start, we divided the work equally. We both conducted research and agreed on the database to use. Together, we decided which feature to include and which to leave out. We each wrote and added our own parts to the report document as well. Since we live close to each other, we were able to collaborate in person most of the time.

Source

Old Dataset:

<https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset/code>

New Dataset:

<https://www.kaggle.com/datasets/tanyamishra260/health-risk-predictor-dataset>